

# Shaft Rope

26/08/2019

# Rope I

- \*There are three common use of steel ropes.
- \*Hoist rope are round stranded, flattened strand, locked coil rope.
- \*Balance rope are non rotating rope.
- \*Guide & Rubbing Rope are half coil rope.
- \*Each rope are with particular construction applicable for each use.

# Rope II

- \*Basic purpose of hoisting rope is to connect conveyance to hoist.
- \*Rope is selected primarily on basis of safety, compatibility, life & cost.
- \*Life of rope for mine winder, including balance rope is defined by mine regulation.
- \*Technically, life of rope depends on number of trip, it will make.
- \*True life of winding rope depends on rope construction, hoist & sheave damnations, type of loading, shaft atmosphere & maintenance.

# Rope III

- \*Cost of initial purchase, maintenance & cost of rope changing, including loss in production are normally considered.
- \*Winding system also has to consider balance rope (counter rope) for friction hoist application, guide ropes (rubbing rope), rope attachment.
- \*Drum winder use round strand lang lay flattened rope.
- \*Friction winder rope is locked coil, flattened strand rope.
- \*Balance rope, non rotating, nylon core
- \*Shaft sinking, non rotating, full locked coil rope. 4

# Rope IV

\*Sling, Round strand rope.

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# Rope Construction I

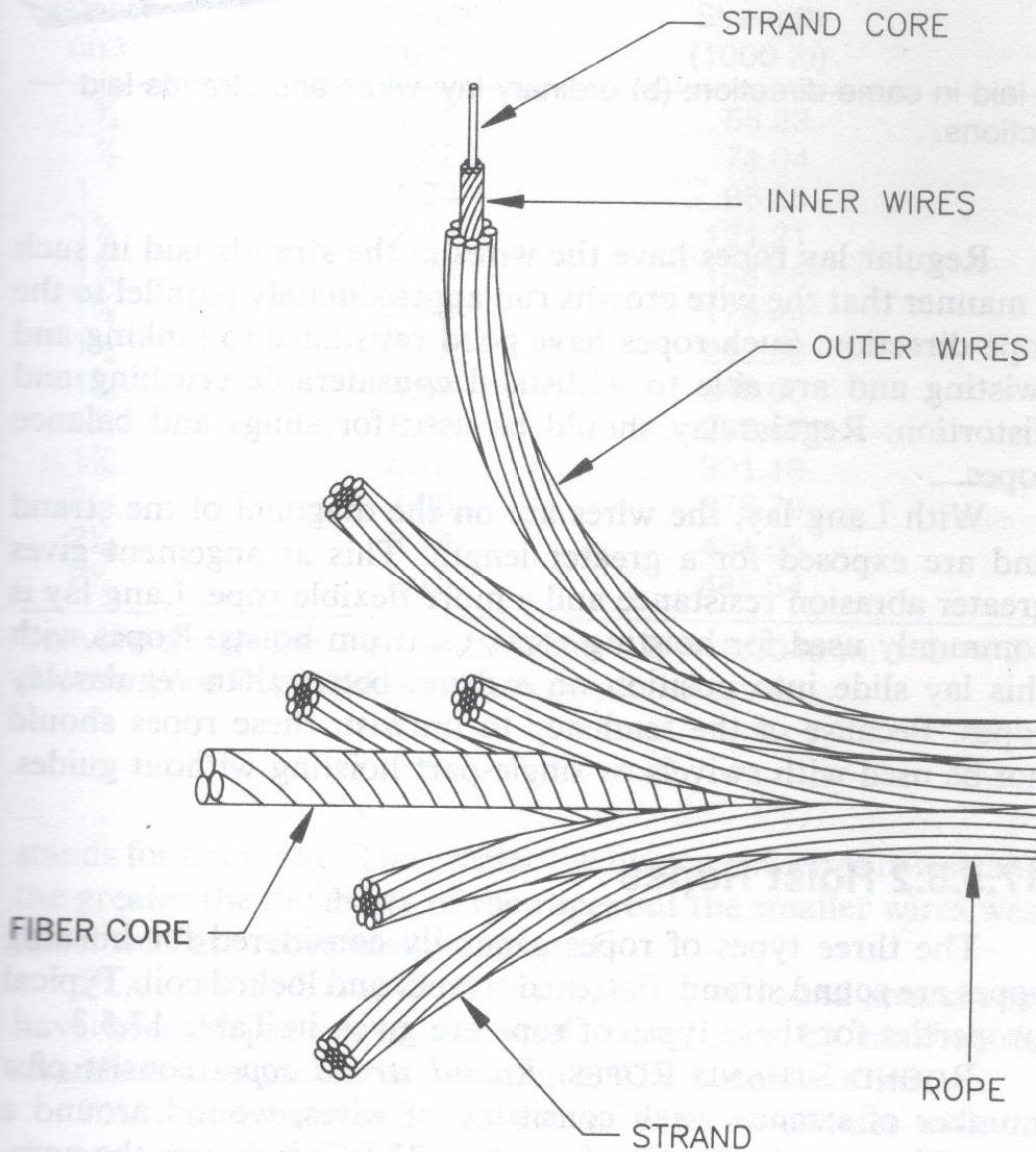
- \*Rope is constructed out of number of individual wires, around a core to form rope.
- \*Some ropes are constructed with out strands by winding consecutive layers of wire around inner layer.
- \*Rope construction consists of
  - \*Wire
  - \*Strands
  - \*Cores
  - \*Lays

# Rope Contraction II

\*By varying these factors, ropes with different characteristic can be constructed.

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# Rope Components I



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Fig. 17.5.10. Rope components (Anon., 1980a).



# Wire I

- \*Wire for manufacturing rope is available with varying strength up to 2500 Mpa.
- \*It is possible to manufacture rope of different strength for same diameter & construction.
- \*As wire strength increases, fatigue life & rope life tends to decrease.
- \*Most common wire used is improved plow grade wire.
- \*Wire is available in different shapes with round, full lock, half lock being most common.

# Wire II

\*Galvanized wire is available for wire rope & should be considered if rope are to be replaced for corrosion.

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# Strands I

- \*Individual ropes are twisted together to form strands.
- \*There are four common type of strands.
- \*Round strands.
- \*Triangular strands
- \*Oval strands
- \*Flat strands

# Cores I

- \*Purpose of core is to take internal compressive stress.
- \*Largest of these stress is due to tendency for rope to flatten when bent around a sheave or thimble under load.
- \*Natural fiber cores (FC) such as sisal & hamp are suitable for most application.
- \*Synthetic fiber core such as nylon & polypropylene are used in conjunction with high strength wire & in corrosive application.

## Cores II

- \*Independent wire rope core (IWRC) are stronger than of fiber cores.
- \*They add to strength of rope & reduce stretch, fatigue life of rope is reduced.

# Lay I

- \*Lay of rope is manner in which wires are twisted into strand & strands are twisted in to ropes.
- \*There are 3 properties used to describe lay, length (pitch), direction (right or left), type (regular or lang).
- \*Two type of lays, regular lays & Lang lay.
- \*Most ropes in mines are Lang lay.
- \*Lay length is distance from one crown to next crown in a particular rope.
- \*Right lay is usually used unless special circumstance requires left lay.

# Lay II

- \*Some drum are designed to take left handed lay ropes & should be checked prior to purchase.
- \*Regular lay ropes have wire in standard layed in such a manner that wire crowns run approximately parallel to rope direction.
- \*Such ropes have good resistance to kinking & twisting.
- \*These rope are able to with stand considerable crushing & distortion.
- \*Regular rope should be used for sling & balance rope.

# Lay III

- \*Lang lay, wires are on diagonal of strand & exposed for greater length.
- \*This arrangement gives greater abrasion resistance & more flexible rope.
- \*Lang lay is commonly used for hoisting purpose on drum hoist.
- \*Ropes with this lay slides in to position on a drum, better than regular lay ropes.
- \*Because of this tendency, these rope should be used with swivels or single point hoisting with out guide.

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# Rope Lay I

MINING ENGINEERING HANDBOOK

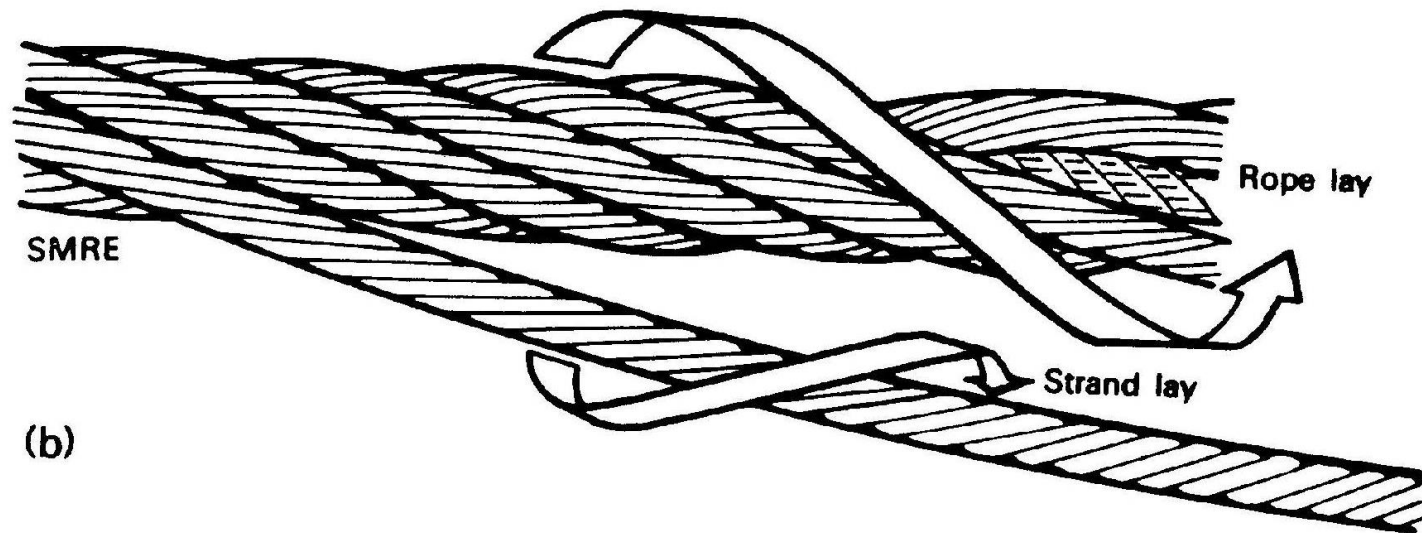
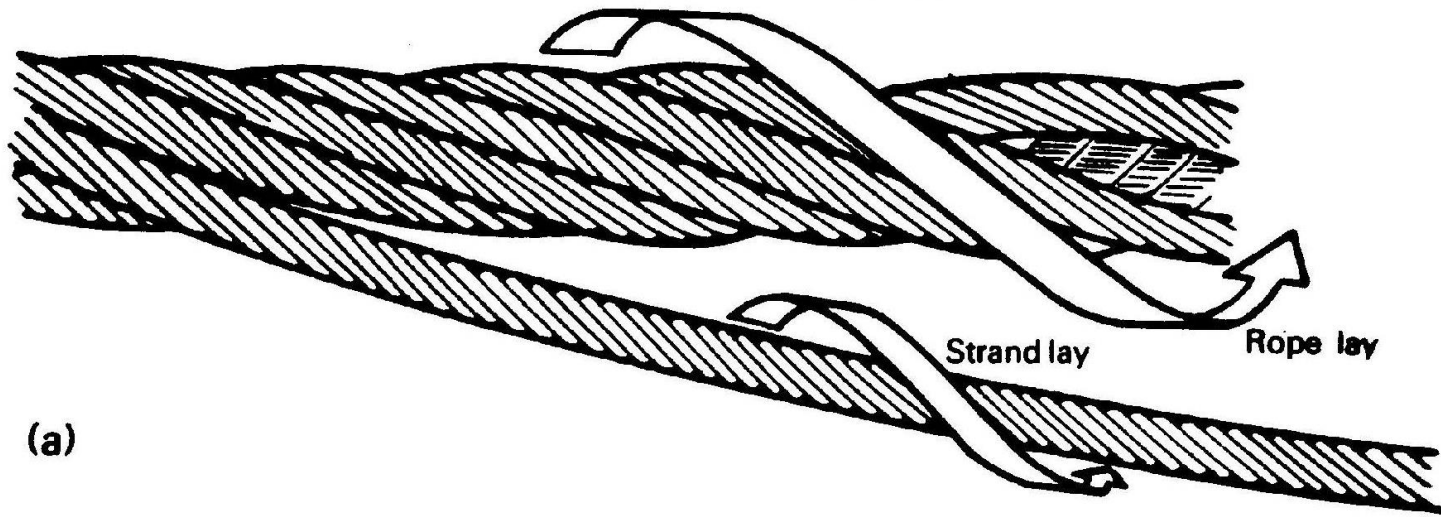


Fig. 17.5.11. Rope lay (Anon., 1980a). (a) Lang lay: wires and strands laid in same direction; (b) ordinary lay: wires and strands laid in opposite directions.

# Direction of Rope Lay I

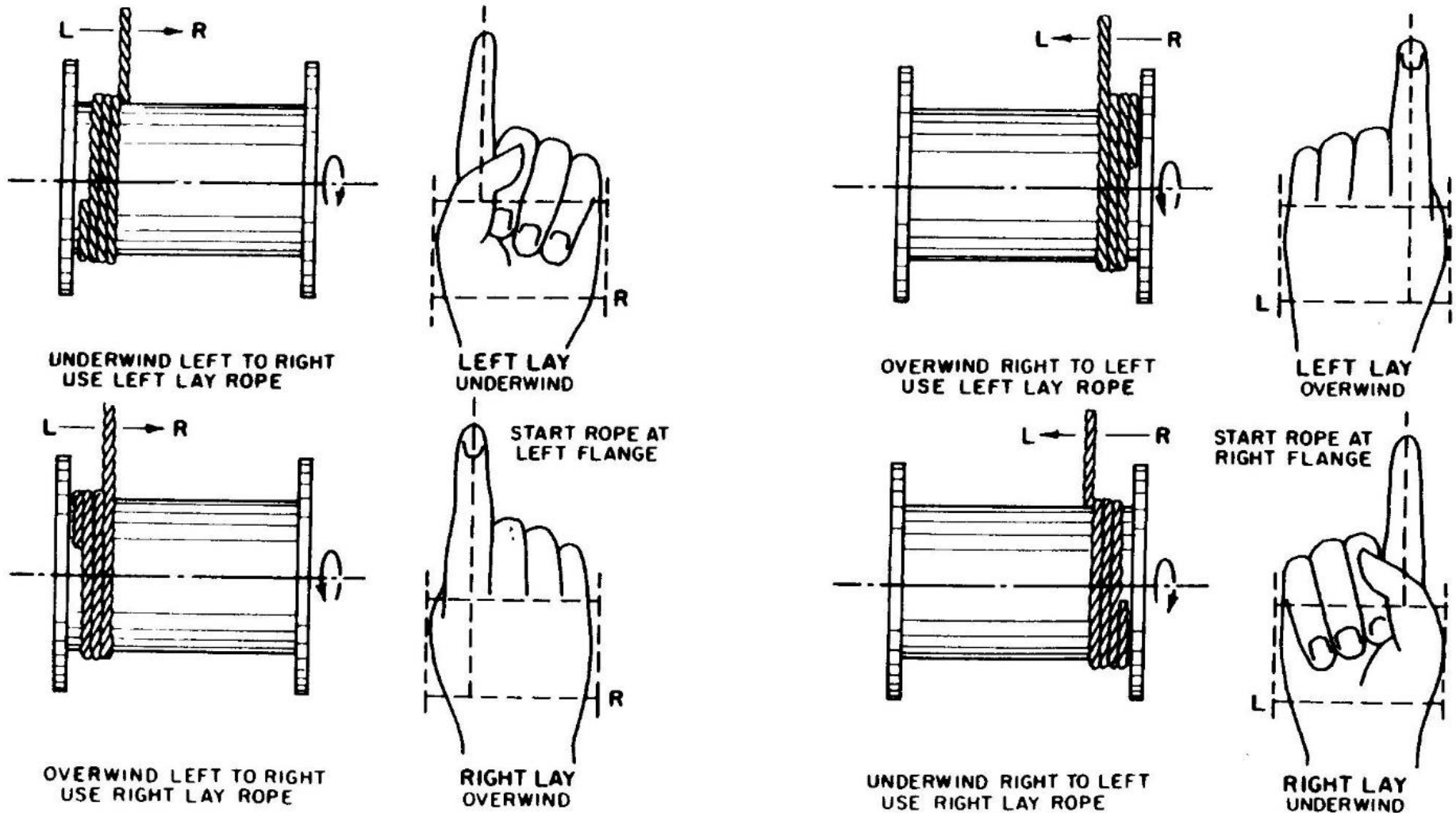


Fig. 17.5.12. Selection of correct direction of lay for a drum hoist ("Wire Rope Users Manual," 2nd ed., Committee of Wire Rope Producers, American Iron and Steel Institute, undated).

# Hoist Rope I

\*Three type of ropes are generally considered for hoisting.

\*Round strand.

\*Flattened strand

\*Lockrd coil

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# Round Strand Rope I

- \*Round strand rope consists of a number of strand, each consisting of wire, wound around a core.
- \*Rope designation (6X27 FC) indicates number of strand (6) & number of wire (27) per strand.
- \*FC stands for fiber core.
- \*Greater number of wire in strand, greater flexibility of rope.
- \*Smaller wire diameter gets damaged faster.
- \*
- \*

# Flattened Strand Rope I

- \*Properties of round strand rope have been improved in flattened strand rope by making strands triangular instead of round.
- \*An increase in number of wires in contact with sheaves & drum increases contact area, thereby reducing pressure & lessening wear of wires, sheaves & drum.
- \*Rope has greater crushing strength & can be used in multi layer winders.
- \*Wire density in rope is higher.
- \*For same diameter, more steel is available.

# Flattened Strand Rope II

\*Rope has higher strength.

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# Full locked Coil Rope I

- \*Centre or core of locked coil rope consists of concentrically laid strand of round wire.
- \*Around these core lies one or more layer of shaped wire, outer layer always being interlocked.
- \*Shape of all shaped wire in locked coil rope depends on rope diameter & its end use.
- \*In locked coil construction, there is no initial twist in rope, so wire does not tend to untwist under load.
- \*Locked coil rope has higher breaking strength than standard rope for equal diameter & same nominal strength grade.

# Full locked Coil Rope II

- \*Because of there smooth external surface, reduction in strength caused by frictional wear on drum or pulley is much lower.
- \*Because of these design, licked coil ropes are less subjected to rotation & stretch then standard rope.

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# Rope Selection I

- \*For initial selection of rope following requirement of rope to be considered.
- \*Strength.
- \*Resistance to being fatigue.
- \*Abrasion resistance.
- \*Resistance to crushing or distortion.
- \*Strength of rope is most important factor to be considered.
- \*For other parameter, balanced approach has to be taken.

# Rope Selection II

- \*Final design criteria shall be to deploy rope which is most economically fulfilling statutory / regulatory criteria.
- \*Hoisting system should be designed to obtain good rope life.
- \*Specification of winder drum, sheave pulley, head frame should be computable to rope to be used in winding system.
- \*Following procedures / systems are to be designed well in advance.

# Rope Selection III

- \*Design rope storage & handling procedure.
- \*Design rope installation Procedure.
- \*Design rope maintenance procedure.
- \*Design initial rope inspection procedure.
- \*

# Rope Selection Drum Winder I

- \*Following items are relevant to drum hoisting.
- \*Factor of safety, as guided by law.
- \*Ratio of drum/ sheave diameter to rope diameter.
- \*Fleet angle.
- \*Factor of safety is taken considering major static load on rope is fully loaded conveyance at deepest point of hoisting.
- \*In Indian condition, drum & rope diameter is 1:100.
- \*Winding system designed, must satisfy these requirement.

# Rope Selection Drum Winder II

- \*Maximum fleet angle permitted in Indian condition is 1.5 degree.
- \*Lower fleet angle provides better rope life.
- \*Higher fleet angle causes piling up of rope.
- \*1.2 degree fleet angle is most comfortable & its provides better life for head sheave, rope etc.

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# Rope Selection Friction Winder I

- \*In friction winder, rope only passes around drum, while friction providing force necessary to move rope.
- \*Friction winder can be tower mounted or ground mounted.
- \*With increase in number of rope for friction winder, diameter of individual rope reduces.
- \*Hoist drum can be of smaller diameter.
- \*This in turn reduces weight of winder, hence cost of winder.
- \*Capital expenditure on winder is lower than that of drum winder of same capacity.

# Rope Selection Friction Winder II

- \*As per regulation, factor of safety requirement for rope remains same as that of drum winder.
- \*For operating friction winder, friction of rope in respect to drum is very important.
- \*Following items are important for selecting proper rope for friction winder.
- \*T1/T2 ratio.
- \*Drum tread.
- \*Deflection sheave.
- \*Different rope loads.
- \*Tail Ropes

# Friction Winder T1/T2 I

- \*T1/T2 ratio is ratio of static rope tension on each side of drum.
- \*T1 is weight of loaded conveyance, plus weight of rope & T2 is weight of unloaded conveyance plus weight of ropes.
- \*Maximum recommended value for particular installation is a function of coefficient of friction of tread material, acceleration & retardation requirement & angle of lap on drum.
- \*A typical maximum value of T1/T2 is 1.5.

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# Friction Winder T1/T2 II

- \*There is also a concern for brake torque application.
- \*Should T1/T2 becomes too high, rope will slip on drum.
- \*Once rope starts slipping over drum, co-efficient of friction drops & control is lost.
- \*There shall be a tread damage due to friction temperature raise & abrasion

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# Friction Winder Drum Tread I

- \*Selection of an appropriate thread material for drum rope groove & tread pressure are important.
- \*As rope passes over drum, load thus strain in rope changes to creep (as opposed it to slip) on drum..
- \*A resilient tread material such as polyurethane tends to flow with changes in strain & reduce creep.
- \*Typically, average tread pressure (sum of rope tension / protected rope area on drum) with locked coil rope is 1800kpa.
- \*With flattened strand rope value is 400kpa

# Friction Winder Drum Tread II

\*Flattened strand & round strand rope on plastic tread will give longer rope life.

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# Friction Winder Deflection Sheave I

- \*Deflection sheave are used when pitch circle diameter of friction drum is greater than center to center distance of compartment.
- \*In addition to positioning hoisting ropes, use of deflection sheaves has advantages of increasing angle of contact of ropes on winder drum, permitting a higher  $T_1/T_2$  ratio before slippage will occur.
- \*Deflection sheave have disadvantages of requirement of additional torque during hoisting cycle, increasing height of head frame & putting reverse bending in to rope, which can reduce life of rope.

# Friction Winder Deflection Sheave I

- \*Position of deflecting sheave is such that rope travel at full speed should take a minimum of 0.5 second to travel from drum to deflection sheave.
- \*This avoids fatigue problem due to reverse bending since it provides requisite time for rope strands & wire to adjust there position.

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# Friction Differential Rope Load I

- \*A great concern with multiple rope friction hoist is equalization of loads between ropes.
- \*Two most common cause of differential loading are a difference in rope length or circumference length of each rope groove on winder drum.
- \*At initial design stage, care should be taken to reduce differential rope loading.
- \*Ratio of rope lengths (skip at dumping point to hoists drum) / (skip at loading pocket to hoist drum) should be greater then 0.015.

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# Friction Differential Rope Load II

- \*Total number of revolution of drum on a trip should be less than 100.
- \*Number of rope should be kept minimum.
- \*There will be less difficulty in maintaining equal tension on a 4 rope winder than 6 rope hoist.
- \*Larger drum diameter will be required for a 4 rope drum having fewer revolutions per trip, lower rope bending stress and lower sensitivity to tension variation,,
- \*There is conflict with rope size & number of ropes.,

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# Friction Differential Rope Load I

\*Rope should have provision of course & fine length adjustment.

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# Friction Winder Speed I

- \*There are no relationship attributed to effect of speed on friction winder.
- \*As speed of hoisting increases, any inadequacies in winding system gets magnificent.

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# Friction Winder Tail Rope I

\*In friction hoisting, use of tail ropes makes it possible to balance head rope weight & achieve a lower peak horse power than drum hoist.

\*Use of tail rope presents some additional problem.

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# Friction Winder Tail Rope Orientation I

- \*Any object falling down a mine shaft is affected by angular velocity of earth & will move towards east wall.
- \*A tail rope behaves in a similar way.
- \*If shaft compartment is on east west, tail rope does not foul with shaft compartment.
- \*In case of north, south compartment, rope collides foul separator at shaft bottom.
- \*While designing shaft, this consideration should be kept.
- \*This action of rope is known as “Coriole” effect.

# Friction Winder Tail Rope Orientation II

“Coriole” effect is a function of hoist speed.

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# Friction Winder Tail Rope Diameter I

- \*A natural loop diameter exist for each type of tail rope diameter depending up on its construction.
- \*This should consider as closely as possible to centre to centre distance of conveyance.
- \*Bending ratio of rope is defined as natural bending divided by rope size.
- \*It is possible to specify different rope construction & number of tail rope to satisfy loop diameter.
- \* Natural loop diameter for tail rope is provided by manufacturer.

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# Friction Winder Tail Rope Diameter I

Rope Construction

Natural Bending Ratio

\* 34X7 non rotating

46:1

\*18X7 non rotating

60:1

\*6X7 Round Strand

90:1

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# Tail Rope Loop Divider I

- \*To avoid mishap & possible mechanical damage through entanglement, balance rope loop are should be well constructed in sump.
- \*This is done by constricting loop divider.
- \*Balance rope should be restricted quite closely in division of their lateral movement.
- \*Early contact with divider between individual rope will provide good control in sump area & also assist in encasing that adequate stability through shaft maintenance.
- \*Longitudinal movement should not overly be restricted.

# Tail Rope Loop Divider II

- \*Divider should be designed to avoid trapping any material from spillage.
- \*Wire rope covered with rubber hose & placed in same configuration has proved most successful in this connection.
- \*Elevation of loop divider for good result is one loop diameter plus 1 meter above bottom of loop.
- \*Some operator place such divider at several elevation in sump area to ensure good control.

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# Tail Rope I

## Tail rope loop divider

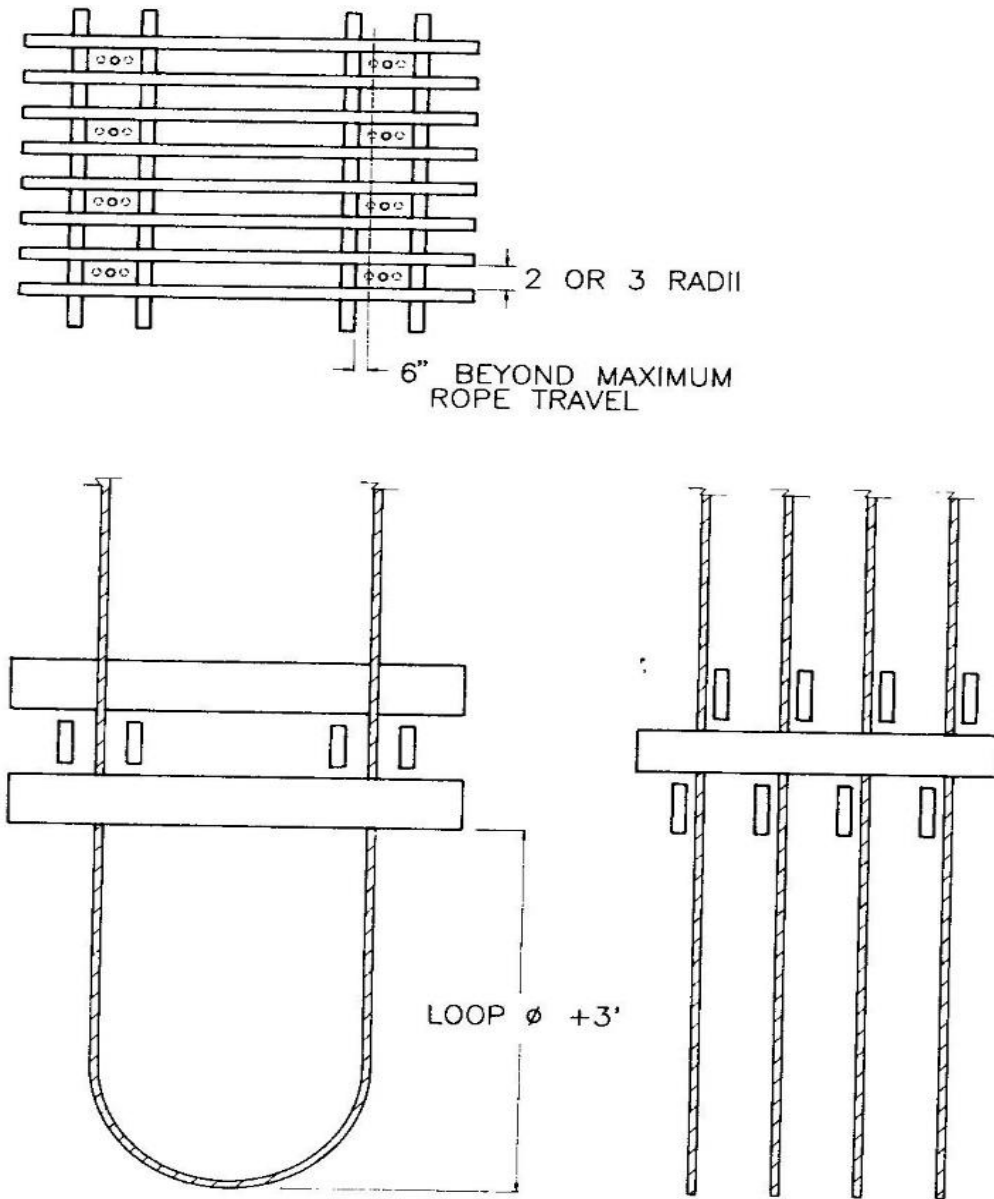


Fig. 17.5.14. Rope loop divider (Delorme, 1977).

# Guide & Rubbing Rope I

- \*Ropes used as guide & rubbing ropes may be locked coil or half locked coil construction.
- \*These ropes are manufactured with core of concentrically laid round wire covered around one or two layer of shaped wires.
- \*Outer layer of locked coil construction is always composed of inter locking wires of Z shaped.
- \*Advantage of locked coil ropes as guide & rubbing rope are provided below.
- \*Smooth surface reduces friction.
- \*Have higher degree of rigidity.

# Guide & Rubbing Rope II

\*Higher breaking strength in relation to their diameter.

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# Rope Stretch I

- \*When skip is loaded at loading pocket rope stretches.
- \*Amount of stretch has static & dynamic components.
- \*Rope act as spring & helps to absorb impact load.
- \*Longer rope, greater stretch, greater amount of energy, that can be absorbed.
- \*This is reason for reducing safety factor for longer rope.
- \*In hoisting system with balanced hoisting using two skip, amount of stretch must be considered, for designing head frame.

# Rope Stretch II

- \*Under normal loading condition one skip will be positioned at loading pocket after skip at head frame has completed dumping.
- \*In case, skip which is at loading point is not loaded with ore, stretch due to loading shall not take place.
- \*Should empty skip is hoisted to surface, it will be above head frame dump ( by amount of rope stretch that did not take place) when lower skip is at loading point.
- \*10 ton pay load skip, 1200m rope length, 38 mm diameter rope would stretch 1. to 1.8m.

# Rope Stretch III

- \*Due to both static & dynamic force, skip moves down during loading.
- \*Unless this is considered in design of loading pocket & skip, considerable spillage could occur.
- \*Static & dynamic rope stretch at loading pocket must be considered in establishing skip travel in shaft bottom.

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# Rope Stretch IV

## Rope Stretch

$$\text{Stretch} = PL/AE$$

P > load in kg

L > length of rope from hoist to loading point m

A > nominal cross section area in mm sq

E > constant for a given rope construction.

\*E is analogous to young modulus.

\*E is determined experimentally by rope manufacturer, taking in consideration of voids between wires in rope & geometric changes in rope under load.

# Rope Stretch V

- \*E value of typical rope.
- \*Full locked coil 99 mpa
- \*Half locked coil 108 mpa
- \*6X27 Flattened Strand 62 mpa
- \*18 X 7 IWRC Sinking 70 mpa
- \*34 X 7 IWRC Sinking 54 mpa
- \*6X19 Round Strand FC 62 mpa
- \*6X19 Round Strand WRC 68 mpa
- \*IWRC = Independent Wire Core (in place of fiber core)



# Rope Stretch VI

- \*Amount of dynamic stretch that takes place is proportional to mass & velocity from impact of rock feeding in to skip..
- \*This factor can be controlled by (to reduce stretch) through loading bin design.
- \*Dynamic stretch is commonly taken as equal to static stretch produced by load weight.

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# Rope Torque I

- \*Standard rope has tendency to untwist.
- \*In drum hoist, rope is fixed at drum & conveyance's tendency to untwist is restrained by guide & shoes.
- \*Consequently torque is developed in rope.
- \*Counter torque exerted by conveyance tends to produce wear on one side of guide.
- \*When disconnected from conveyance, torque is released & dangerous situation develops if end of rope is not restrained & spin out of control.
- \*Some time deliberately letting torque out of rope when rope sample is taken for testing.

# Rope Torque II

- \*Some experiments shows that , allowing a rope to release torque, ultimate strength is lost.
- \*Torque in rope depends on diameter of rope, rope construction. Load on rope.

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# Rope Harmonics & Vibration I

- \*Vibration of rope can be observed at shaft collar for friction winder.
- \*In case of drum winder an rope whipe between winder & head sheave.
- \*Causes of this vibration are.
- \*Jerking in operation, such as uneven breaking.
- \*Horizontal movement of rope at cross over point on drum
- \*These vibration can not be avoided & normally do not cause problem.

# Rope Harmonics & Vibration II

- \*In case, where frequency of vibration is a harmonics of natural frequency of winder rope, then severe damage & whipping shall occur.
- \*In case where flat span of rope between winder & head sheave is long & flat.
- \*To avoid unnecessary vibration & whipping following points are to be considered.
- \*Vertical angle of rope should be as close as 45 degree.

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# Rope Harmonics & Vibration III

- \*Slope distance along rope, between tangent point at drum & sheave, should not be an even multiply of drum circumference.
- \*Hoist design should analysis harmonics behavior of rope & select type of drum grooving, which will avoid harmonic vibration.
- \*When unaccepted rope movement do occur, it is necessary to reduce winding speed, pay load or to install intermediate tower with idler sheave between winder & head fame.

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# Rope Sag I

- \*Rope between around mounted winder & a head frame sags & takes form of catenary.
- \*With heavy rope & light load, sag can be significant & cause problem at sheave deck & hoist house wall.
- \*Such situation can occur during sinking a deep shaft, when empty bucket at dump.

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# Rope Testing I

- \*Manufacturer are to provide certificate to purchaser to satisfy that rope should meet requisite like nominal breaking load.
- \*Certificates includes a test providing full particulars of wires used to manufacture rope.
- \*Test to destruction by an independent laboratory is done.

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# Rope Storage I

- \*All rope should be stored indoor, shielded from weather, corrosive fumes & excessive heat.
- \*Excessive heat dries out lubricants of fiber core.
- \*Rope should be stored in original rope reel.
- \*Rope reel should be turned periodically to prevent collection at bottom of reel.

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# Rope Installation I

- \*Rope should be wound on to drum under load.
- \*Direction of coiling should be determined by rule of thumb.
- \*Individual turn of rope on drum should be tamped together using a copper hammer to avoid gaps between coils.
- \*Before cutting ropes, rope should be binded at several places with help of GI weir & soldered at sides.
- \*This binding (servings) should not be on portion of rope which runs over drum or sheave pulley.

# Drum Winder Rope Installation I

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# Friction Winder Rope Installation I

- \*There are several process for friction winder rope change.
- \*Exact process of rope changing varies for unit to unit, because of experience, past practice & specific mine condition.
- \*There are several method to change rope.

## Changing station at mid of shaft

- \*A rope changing station is created at mid way in shaft.
- \*Normally, a shaft landing level at mid of shaft serves as temporary rope changing station.

# Friction Winder Rope Installation II

- \*Main rope (head rope) are changed by chairing both conveyance, skip / cage & counter weight at mid of shaft.
- \*New rope reels are lowered at mid shaft well in advance by cage or by separate tugger hoist from surface.
- \*Disconnect both end of main rope.
- \*Connect one end of old rope with one end of new rope.
- \*Opposite end of old rope is connected to empty reel.

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# Friction Winder Rope Installation III

- \* All main ropes are changed one by one by rotating friction drum winder & thus lifting new rope in position.
- \* Old ropes are wound on empty reels.
- \* After new ropes has been hoisted in to position & both of new ropes are connected to conveyance.

## By rope winch at surface

- \* A portable rope winch located on surface removes all main rope (head rope) one at a time, winding them on empty reel at surface.

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# Friction Winder Rope Installation IV

- \*New main ropes are lowered by these winch & installed on conveyance.
- \*During this period one conveyance is located at top of shaft & another at bottom of shaft.

## Changing Tail (balance ) rope.

- \*Tail ropes are installed from bottom of shaft.
- \*Locate no 1 conveyance at bottom of shaft.
- \*Locate no 2 conveyance at top of shaft.
- \*Release rope clamp on bottom of no 1 conveyance & start reeling rope to be removed on a spare reel.

# Friction Winder Rope Installation V

- \*Lower no 2 conveyance & remove rope clamp on bottom of no 2 conveyance & complete winding up end of rope removed.
- \*Install new tail rope with reverse procedure.
- \*New tail rope will be hoisted in position.
- \*Final method of rope changes depends up on amount of room available on surface, orientation of hoist rope, availability of soace in head frame & personal preference.

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# Rope Operation I

- \*Mine rope is vital for winder operation.
- \*Care must be taken to operate & maintenance of rope.
- \*Drum & sheave groove counters are correcte & maintained.
- \*Sheave groove, if found damaged or deformed, sheave should be changed to prevent rope damage.
- \*Filler, starter & raiser strip used on drum, where necessary.
- \*Rope cross over on drum should be prevented.
- \*Shock loading & over loading should be avoided.

# Rope Operation II

- \*Vibration stresses are to be kept as minimum as possible.
- \*Peening & abrasive action are minimized.
- \*Rope cut back are appropriately completed.
- \*Conveyance end attachment are refastened at regular interval.
- \*Rope is to be lubricated at regular interval with correct lubricants.
- \*

# Winder Maintenance Rope Related I

- \*Maintenance of equipment that comes in contact with rope is important & should not be over looked during design or operation phase.
- \*Tight shave & drum groove will pinch rope, restraining wire & strand movement.
- \*Incorrectly designed rope cross over on drum hoist will crush or flattened rope, induce core to pop & severally scrub wires.
- \*This can be prevented through correctly designed filler, starter & raised strip on drum flange.

# Winder Maintenance Rope Related II

- \*Faulty bearing can cause vibration stress, which in turn causes fatigue fracture that are most likely to concentrate at end attachment.
- \*Corrugated deflection sheaves, drum & roller can cause high unit pressure, aggravating wear, excessive vibration & early wire fatigue,
- \*Correct measuring & trimming of tread diameter will prevent this problem.

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# Rope Lubrication I

- \*Atmospheric moisture, shaft water, chemical fumes & few of agencies that will cause corrosion.
- \*This frequently lead to wire breakage & premature removal of miner hoist rope.
- \*In mine with very wet & corrosive shafts, it is distinctly advantageous to apply a lubricant that will penetrate interior of rope that contains a corrosion inhibitor.
- \*Lubricate with water displacing properties are also available & useful at end attachment.

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# Rope Inspection I

- \*In order to achieve optimum rope performance, an inspection procedure that requires finding to be recorded, fully diagnosed & translated in to appropriate action should be integrated.
- \*Marked reduction in diameter may be due to a loss of rope elasticity, excessive abrasion or corrosion.
- \*Broken weir may be due to wear out sheave or drum groove, tight over sized sheave or gappy coiling.
- \*“Kinking” & “Bird caging” of working rope generally results from a sudden release of load.

# Rope Installation II

- \*Kinking is snarling back of a rope on itself in form of loop, resulting a bend in rope when straitened out.
- \*Bird caging is applied to the spring of stretch away from core or inner strands.
- \*Through evaluation of all facts obtained through routine inspection, examination, test & compliance with statutory mining act in need to decide how early a rope should be discarded for sake of safety & how late it be removed for sake of economy.

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# Rope Attachment I

- \*Wire rope attachments are as important as rope to which they are fastened.
- \*Selection of rope attachment should be based on an understanding of how they affects ultimate rope survivability & efficiency.
- \*There are two ways to attach some thing to a wire rope for winding purpose.
- \*Loop can be made by splicing rope to it self (not allowed in India)
- \*Loop can also be made by using a clamp or wedges.

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# Rope Attachment II

\*Some thing can be attached to rope by pouring a liquid material such as molten zinc, white metal, bt wedging arrangement.

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# Rope Attachment Factor of Safety I

- \*It is required to design rope attachment stronger than rope.
- \*Rope should fail before attachment.
- \*When an attachment is made to a rope, attachment commonly weakens rope locally.
- \*Efficiency of an attachment is expressed as percentage of strength left in rope after installation of attachment.

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# Cappel Attachment I

- \*Cappels are rope fitting attached directly to rope.
- \*This type of fitting consists of a pair of wedges grooved to suit particular rope diameter & interlocked to ensure complimentary movement.
- \*A number of bands driven over diverging exterior surface of cappel limited provides internal compressive force to ensure that wedge grip rope.
- \*Limbs internal surface been machined to fit external surface of wedge.
- \*A safety block is fastened on end of rope, protruding beyond bottom of wedges.

# Thimble Attachment I

- \*Thimble type of attachment is used along with U bolt camp.
- \*A loop is formed in rope & thimble placed with in this loop.
- \*Free end of rope is clamped to long end to secure thimble.
- \*Thimbles are made of cast steel clamps, shackles & pin are made of 1.5% manganese steel.

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# Swivel Attachment I

- \*When winding with balance rope, lay length tends to shortened as each conveyance in turn approaches bottom landing.
- \*Swivels are provided in order to allow rope to spin & regain there normal length.
- \*Shaft condition & landing influences type of swivel used.

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# Attachment Maintenance I

- \*All type of attachment be installed as recommended by winder manufactures.
- \*Winder manufacture's recommendation for correct number of clips, amount of turn back, clip positions & correct torque should be followed closely.
- \*In case of wedge type attachment, care should be taken to properly clean rope surface.
- \*Frequent application of cement dust & subsequent steel brushing will remove all surface free from lubricates.
- \*All type of attachment be inspected both visually & non destructively on regular basis.

# Reference I

\*SME Mining Engineering Handbook, 2<sup>nd</sup> Edition,  
Volume 2, Page 1661 to 1671